

Given: $\frac{d^2 u}{dx^2} - 2u = f(x)$, $0 < x < 1$ — (1)

$f(x) = 4x^2 - 2x - 4$ — (2)

(i) Dirichlet Boundary Conditions.

$u = 0$ at $x = 0$

$u = -1$ at $x = 1$

(ii) Neumann BC

$u = 0$ at $x = 0$

$\frac{du}{dx} = -3$ at $x = 1$

Exact Solⁿ $u = -2x^2 + x$

(I) FINITE DIFFERENCE METHOD (FDM)

$h = \frac{1}{3}$

u_1	u_2	u_3	u_4
0	$\frac{1}{3}$	$\frac{2}{3}$	1
x_1	x_2	x_3	x_4

The 1D domain is divided into 3 elements, showing 4 global nodes.

$$u_i'' = \frac{d^2 u}{dx^2} \Big|_{x_i} = \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}$$

From (1)

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} - 2u_i = f(x_i)$$

$$u_{i+1} - (2 + 2h^2)u_i + u_{i-1} = h^2 f(x_i) \quad \text{--- (3)}$$

From ②

$$\bullet f(x_1) = f(0) = -4$$

$$\bullet f(x_2) = f\left(\frac{1}{3}\right) = 4\left(\frac{1}{9}\right) - \frac{2}{3} - 4 = -\frac{38}{9}$$

$$\bullet f(x_3) = f\left(\frac{2}{3}\right) = 4\left(\frac{4}{9}\right) - \frac{2 \times 2}{9} - 4 = -\frac{32}{9}$$

$$\bullet f(x_4) = f(1) = 4 - 2 - 4 = -2$$

(i) Writelet $[u_1=0, u_4=-1]$, $\{ \text{unknowns: } u_2, u_3 \}$

From ③

• Node 2:

$$u_3 - \frac{20}{9}u_2 + u_1 = \frac{1}{9}\left(-\frac{38}{9}\right) - \textcircled{1a}$$

$$-\frac{20}{9}u_2 + u_3 = -\frac{38}{81} - \textcircled{4}$$

• Node 3

$$u_4 - \frac{20}{9}u_3 + u_2 = \frac{1}{9}\left(-\frac{32}{9}\right)$$

$$u_2 - \frac{20}{9}u_3 = -\frac{32}{81} + 1$$

$$u_2 - \frac{20}{9}u_3 = \frac{49}{81} - \textcircled{5}$$

From ④ & ⑤

$$\begin{bmatrix} -20/9 & 1 \\ 1 & -20/9 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -38/81 \\ 49/81 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + \frac{9}{20} R_1$$

$$\begin{bmatrix} -20/9 & 1 \\ 0 & -1.772 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -38/81 \\ 0.394 \end{bmatrix}$$

$$u_2 = 0.111 = 1/9$$

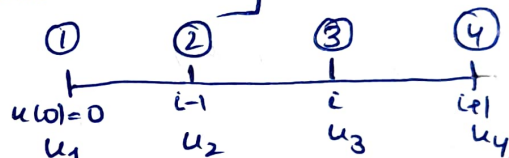
$$u_3 = -0.222 = -2/9$$

$\therefore$ FDM (direct) solⁿ:

$$u_1 = 0, u_2 = \frac{1}{9}, u_3 = -\frac{2}{9}, u_4 = -1$$

(ii) Neumann solⁿ $[u_1 = 0, \frac{du}{dx}(1) = -3]$

Croston Point Method.



At node 3:

$$\frac{du}{dx}(x_3) = -3 \quad (\text{using 2nd order backward difference})$$

$$\frac{du}{dx} \Big|_{x_3} \approx \frac{3u_4 - 4u_3 + u_2}{\Delta x} = -3 \Rightarrow u_2 - 4u_3 + 3u_4 = -2 \quad (7)$$

At node 2: (4) eqⁿ

At node 3: (3) eqⁿ

$$u_4 - \frac{20}{9} u_3 + u_2 = -\frac{32}{81} \quad (6)$$

$$-2 + u_4 - \frac{20}{9} u_3 = -\frac{32}{81}$$

$$u_4 - \frac{20}{9} u_3 = -\frac{32}{81} \quad (5)$$

from (4), (6) & (7)

$$\begin{bmatrix} -\frac{20}{9} & 1 & 1 \\ 1 & -\frac{20}{9} & 1 \\ 1 & -4 & 3 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -\frac{38}{81} \\ -\frac{32}{81} \\ -2 \end{bmatrix}$$

$$u_2 = \frac{1}{9}, \quad u_3 = -\frac{2}{9}, \quad u_4 = -1,$$

$\therefore$ FDM (Neumann) solⁿ

$$u_1 = 0, \quad u_2 = \frac{1}{9}, \quad u_3 = -\frac{2}{9}, \quad u_4 = -1$$

Exact solⁿ (given)

$$u(x_1) = 0, \quad u(x_2) = 0.111\ldots, \quad u(x_3) = -0.222, \quad u(x_4) = -1$$

FINITE ELEMENT METHOD

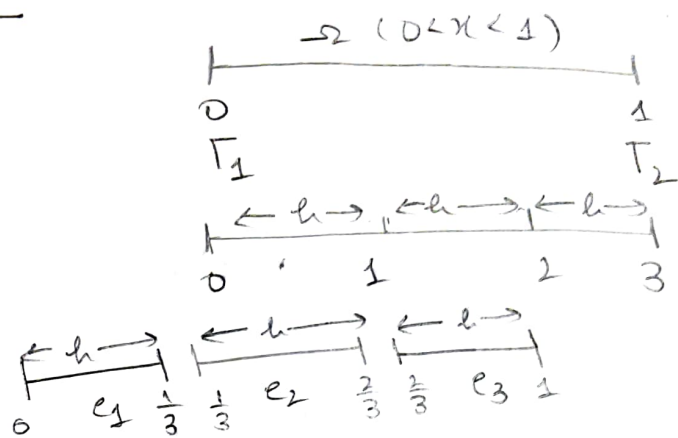
• Global domain

$(\alpha = 0, 1, 2, 3)$

$$e_i = \frac{1}{3}$$

• Local Domain

$$(N, M) = 1, 2$$



• Galerkin Formulation & the Weak form.

① Residual ($R^{(e)}$):

$$R^{(e)} = \frac{d^2 u^{(e)}}{dx^2} - 2u^{(e)} - f(x)$$

② Orthogonality $\int_e \phi_m^{(e)} R^{(e)} dx = 0$

Weak form:

$$-\int_e \left(\frac{d\phi_m}{dx} \cdot \frac{du}{dx} + 2\phi_m u \right) dx = \int_e \phi_m f(x) dx - \left[\phi_m \frac{du}{dx} \right]$$

- Stiffness matrix $[K^{(e)}]$

$$[K^{(e)}] = - \underbrace{\int_e \frac{d\phi^T}{dx} \cdot \frac{d\phi}{dx} dx}_{\text{diff.}} - \underbrace{\int_e 2\phi^T \phi dx}_{\text{mass}} \quad \text{--- ②}$$

$$- [K_{diff}^{(e)}]$$

$$= - \int_0^h \frac{d\phi^T}{dx} \cdot \frac{d\phi}{dx} dx$$

$$= - \int_0^h \begin{bmatrix} \left(\frac{d\phi_1}{dx}\right)^2 & \left(\frac{d\phi_1}{dx}\right)\left(\frac{d\phi_2}{dx}\right) \\ \left(\frac{d\phi_2}{dx}\right)\left(\frac{d\phi_1}{dx}\right) & \left(\frac{d\phi_2}{dx}\right)^2 \end{bmatrix} dx$$

• Derivatives of Linear shape functions.

$$- \phi_1(x) = 1 - \frac{x}{h} = 1 - 3x, \quad \frac{d\phi_1}{dx} = -3$$

$$- \phi_2(x) = \frac{x}{h} = 3x, \quad \frac{d\phi_2}{dx} = 3$$

$$- a_{11} = \int_0^h \left(\frac{d\phi_1}{dx}\right)^2 dx = -3$$

$$- a_{12} = \int_0^h \frac{d\phi_1}{dx} \cdot \frac{d\phi_2}{dx} dx = \int_0^h \left(-\frac{1}{h}\right)\left(\frac{1}{h}\right) dx = \frac{1}{h} = 3 = a_{21}$$

$$- a_{22} = \int_0^h \left(\frac{d\phi_2}{dx}\right)^2 dx = \int_0^h \frac{1}{h^2} dx = -\frac{1}{h} = -3$$

$$- [K_{diff}^{(e)}] = -\frac{1}{h} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{--- (9)}$$

$$- [K^{(e)}]_{\text{mass}}$$

$$= - \int_0^h 2 \phi^T \phi dx = -2 \int_0^h \begin{bmatrix} \phi_1^2 & \phi_1 \phi_2 \\ \phi_2 \phi_1 & \phi_2^2 \end{bmatrix} dx$$

$$a_{11} = -2 \int_0^h \phi_1^2 dx = -2 \int_0^h \left(1 - \frac{x}{h}\right)^2 dx = -\frac{2h}{3}$$

$$a_{21} = a_{12} = -2 \int_0^h \phi_1 \phi_2 dx = -2 \int_0^h \left(1 - \frac{x}{h}\right) \left(\frac{x}{h}\right) dx = -\frac{h}{3}$$

$$a_{22} = -2 \int_0^h \phi_2^2 dx = -2 \int_0^h \left(\frac{x}{h}\right)^2 dx = -\frac{2h}{3}$$

$$\therefore [K^{(e)}] = -\frac{2h}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{--- (9)}$$

Substituting (9) & (10) in (8) $\left(h = \frac{1}{3}\right)$

$$[K^{(e)}] = -3 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} - \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$[K^{(e)}] = \frac{1}{9} \begin{pmatrix} -29 & 26 \\ 26 & -29 \end{pmatrix}$$

• For element 1 (Nodes 0-1)

$$[K] = \frac{1}{9} \begin{pmatrix} -29 & 26 & 0 & 0 \\ 26 & -29 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

• Adding element-2 (Nodes 1-2)

$$K_{11} = K_{22}^{(1)} + K_{11}^{(2)} = -\frac{29}{9} + \left(-\frac{29}{9}\right) = -\frac{58}{9}$$

$$[K] = \frac{1}{9} \begin{pmatrix} -29 & 26 & 0 & 0 \\ 26 & -58 & 26 & 0 \\ 0 & 26 & -29 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

• Adding element 3 (Nodes 2-3)

$$K_{22} = \left(-\frac{29}{9}\right) + K_{11}^{(3)} = -\frac{29}{9} + \left(-\frac{29}{9}\right) = -\frac{58}{9}$$

$$[K] = \frac{1}{9} \begin{pmatrix} -29 & 26 & 0 & 0 \\ 26 & -58 & 26 & 0 \\ 0 & 26 & -58 & 26 \\ 0 & 0 & 26 & -29 \end{pmatrix}$$

$$[K][U] = [F]$$

- Source vector $[F]$: $\int_e \phi_N^{(e)} f(x) dx$

- $\phi_1(x) = 1-3x$
- $\phi_2(x) = 3x$
- $f(x) = 4x^2 - 2x - 4$

→ Element 1 (Nodes 1-2, from 0 to $\frac{1}{3}$)

$$\begin{aligned} F_1^{(1)} &= \int_0^{\frac{1}{3}} (1-3x)(4x^2-2x-4) dx = \int_0^{\frac{1}{3}} (12x^3 + 10x^2 + 10x - 4) dx \\ &= -\frac{56}{81} \end{aligned}$$

$$\begin{aligned} F_2^{(1)} &= \int_0^{\frac{1}{3}} (3x)(4x^2-2x-4) dx \\ &= \int_0^{\frac{1}{3}} (12x^3 - 6x^2 - 12x) dx = -\frac{57}{81} \end{aligned}$$

• Element 2 (Nodes 2-3, from $\frac{1}{3}$ to $\frac{2}{3}$)

Coordinate Transform : $x_{\text{new}} = \frac{1}{3} + x$

$$\therefore f(x_{\text{new}}) = 4 \left(\frac{1}{3} + x \right)^2 - 2 \left(\frac{1}{3} + x \right) - 4$$
$$= 4x^2 + \frac{2}{3}x - \frac{38}{9}$$

$$\bullet F_1^{(2)} = \int_0^{1/3} (1-3x) \left(4x^2 + \frac{2}{3}x - \frac{38}{9} \right) dx = -\frac{55}{81}$$

$$\bullet F_2^{(2)} = \int_0^{1/3} (3x) \left(4x^2 + \frac{2}{3}x - \frac{38}{9} \right) dx = -\frac{52}{81}$$

• Element 3 (Nodes 3-4, from $\frac{2}{3}$ to 1)

• Coordinate Transform : $x_{\text{new}} = \frac{2}{3} + x$

$$f(x_{\text{new}}) = 4 \left(\frac{2}{3} + x \right)^2 - 2 \left(\frac{2}{3} + x \right) - 4$$
$$= 4x^2 + \frac{10}{3}x - \frac{32}{9}$$

$$\bullet F_1^{(3)} = \int_0^{1/3} (1-3x) \left(4x^2 + \frac{10}{3}x - \frac{32}{9} \right) dx = -\frac{42}{81}$$

$$\bullet F_2^{(3)} = \int_0^{1/3} (3x) \left(4x^2 + \frac{10}{3}x - \frac{32}{9} \right) dx = -\frac{35}{81}$$

Global Source Vector.

$$\bullet f_{g,1} = F_1^{(1)} = -\frac{56}{81}$$

$$\bullet f_{g,2} = F_2^{(1)} + F_1^{(2)} = -\frac{57}{81} + \left(-\frac{55}{81} \right) = -\frac{112}{81}$$

$$\bullet f_{g,3} = F_2^{(2)} + F_1^{(3)} = -\frac{52}{81} + \left(-\frac{42}{81} \right) = -\frac{94}{81}$$

$$\bullet f_{g,4} = F_2^{(3)} = -\frac{35}{81}$$

$$\therefore F_{global} = \frac{1}{81} \begin{bmatrix} -56 \\ -112 \\ 94 \\ -35 \end{bmatrix}$$

(i) Dirichlet

$$[K][U] = [F]$$

$$\frac{1}{9} \begin{pmatrix} -29 & 26 & 0 & 0 \\ 26 & -58 & 26 & 0 \\ 0 & 26 & -58 & 26 \\ 0 & 0 & 26 & -29 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \frac{1}{81} \begin{pmatrix} -56 \\ -112 \\ 94 \\ -35 \end{pmatrix}$$

$$u_1 = 0 \quad \& \quad u_4 = -1.$$

$$\frac{1}{9} \begin{pmatrix} 58 & -26 \\ -26 & 58 \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 112/81 \\ \frac{94}{81} - (-\frac{26}{9})(-1) \end{pmatrix}$$

$$\frac{1}{9} \begin{pmatrix} 58 & -26 \\ -26 & 58 \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 112/81 \\ -138/81 \end{pmatrix}$$

$$u_2 = 0.1202$$

$$u_3 = -0.2105.$$

$\therefore$ Dirichlet (FEM) solⁿ:

$$u_1 = 0, \quad u_2 = 0.1202, \quad u_3 = -0.2105, \quad u_4 = -1$$

(ii) Neumann

Neumann boundary vector ($G_N^{(e)}$)

$$G_N^{(e)} = \phi_N^{(e)} \frac{du}{dx} \Big|_0^L$$

$$G_3 = - \left(\phi_3(1) \frac{du}{dx}(1) - \phi_3(0) \frac{du}{dx}(0) \right)$$

$$= - (1 \times -3 - 0) = 3$$

Global neumann vector $\{G\} = \begin{Bmatrix} 0 \\ 0 \\ 3 \end{Bmatrix}$

$$\therefore [K]\{U\} = \{F\} + \{G\}$$

$$\frac{1}{9} \begin{pmatrix} -58 & 26 & 0 \\ 26 & -58 & 26 \\ 0 & 26 & -29 \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \\ u_4 \end{pmatrix} = \frac{1}{81} \begin{pmatrix} -112 \\ -94 \\ -35 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

Solving, we get

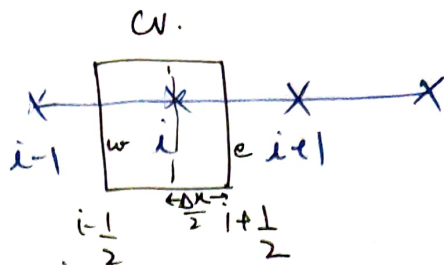
$$u_1 = u_2 \approx 0.1199, u_3 \approx -0.1919, u_4 \approx -0.9515.$$

$\therefore$ Neumann (PEM) Q_1^N :

$$u_1 = 0, u_2 = 0.1199, u_3 = -0.1919, u_4 = -0.9515.$$

Finite Volume Method.

From ①



$$\int_{w_e}^e \left(\frac{du}{dx^2} - 2u \right) dx = \int_{w_e}^e f(x) dx$$

$\left\{ \begin{array}{l} e \text{ at } i + \frac{h}{2} \\ w \text{ at } i - \frac{h}{2} \end{array} \right\}$

$$\int_{w_e}^e \frac{d}{dx} \left(\frac{du}{dx} \right) dx - \int_{w_e}^e 2u dx = \int_{w_e}^e f(x) dx \quad \text{--- (11)}$$

Breaking down:

$$\int_{w_e}^e \frac{d}{dx} \left(\frac{du}{dx} \right) dx = \left(\frac{du}{dx} \right)_e - \left(\frac{du}{dx} \right)_{w_e}$$

$$\left(\frac{u_{i+1} - u_i}{h} \right) - \left(\frac{u_i - u_{i-1}}{h} \right) = \frac{u_{i-1} - 2u_i + u_{i+1}}{h^2}$$

$$\int_{w_e}^e 2u dx = 2u_i h$$

$$\int_{w_e}^e f(x) dx = \int_{i-1/2}^{i+1/2} (ux^2 - 2x - 4) dx = \frac{4}{3} \left[x^3 \right]_{1-\frac{h}{2}}^{1+\frac{h}{2}} - \left[x^2 - 4x \right]_{1-\frac{h}{2}}^{1+\frac{h}{2}}$$

$$\frac{1+\frac{h}{2}}{2} = \frac{1+\frac{h}{2}}{2}$$

$\therefore$ Eqⁿ ⑪ becomes.

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} - 2u_i h = \int_{w_e}^e f(x) dx \quad \text{--- (12)}$$

Dirichlet Boundary Conditions.

($u_1=0$, $u_4=-1$, unknowns: u_2 & u_3)

From (2)

$$\frac{u_{i+1}-u_i}{h} - \frac{u_i-u_{i-1}}{h} - 2u_i h = \int_{x_i-\frac{h}{2}}^{x_i+\frac{h}{2}} f(x) dx$$

At Node 2: ($x_2 = \frac{1}{3}$) CV is from $x: \frac{1}{6} \rightarrow \frac{1}{2}$

$$\text{RHS: } \int_{\frac{1}{6}}^{\frac{1}{2}} (4x^2 - 2x - 4) dx = -\frac{113}{81}$$

$$\text{LHS: } \frac{u_1 - 2u_2 + u_3}{h} - 2u_2 h$$

$$\therefore 3u_1 - 6u_2 - \frac{2}{3}u_2 = -\frac{113}{81} \Rightarrow -\frac{20}{3}u_2 + 3u_3 = -\frac{113}{81} \quad (13)$$

At Node 3 ($x_3 = \frac{2}{3}$) CV is from $x: \frac{1}{2} \rightarrow \frac{5}{6}$

$$\text{RHS: } \int_{\frac{1}{2}}^{\frac{5}{6}} (4x^2 - 2x - 4) dx = -\frac{95}{81}$$

$$\text{LHS: } \frac{u_4 - 2u_3 + u_2}{h} - 2u_3 h = -\frac{95}{81}$$

$$3u_2 - \frac{20}{3}u_3 = +\frac{148}{81} \quad (14)$$

From eqⁿ 13 & 14.

$$\begin{bmatrix} -\frac{20}{3} & 3 \\ 3 & -\frac{20}{3} \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -\frac{113}{81} \\ \frac{148}{81} \end{bmatrix}$$

$$u_2 \approx 0.1077$$

$$u_3 \approx -0.2274.$$

$\therefore$ Dirichlet (FEM) solⁿ

$$u_1 = 0, \quad u_2 = 0.1077, \quad u_3 = -0.2274, \quad u_4 = -1.$$

(ii) Neumann BC. $[u_1 = 0, u'(1) = -3]$
unknowns: u_2, u_3, u_4

• Node 2:

$$-\frac{20}{3}u_2 + 3u_3 = -\frac{113}{81} \quad (\text{from 13})$$

$$\text{• Node 3: } 3u_2 - \frac{20}{3}u_3 + 3u_4 = -\frac{95}{81} \quad - (14)$$

$$\text{• Node 4: } 3u_3 - \frac{10}{3}u_4 = \frac{8}{3} \quad (15) \quad (\text{from 15a})$$

$$\begin{bmatrix} -\frac{20}{3} & 3 & 0 \\ 3 & -\frac{20}{3} & 3 \\ 0 & 3 & -\frac{10}{3} \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -113/81 \\ -95/81 \\ 8/3 \end{bmatrix}$$

$$u_2 \approx 0.1190, \quad u_3 \approx -0.1983, \quad u_4 \approx -0.9543.$$

$\Rightarrow$ Neumann FEM solⁿ

$$\text{let } u_1 = 0, \quad u_2 \approx 0.1190, \quad u_3 \approx -0.1983, \quad u_4 \approx -0.9543$$

(ii) Neumann

$$\left(\frac{du}{dx}\right)_e - \left(\frac{du}{dx}\right)_w - \int_w^e 2u dx = \int_w^e f(x) dx$$

$$\cdot \left(\frac{du}{dx}\right)_e = -3 \quad , \quad \cdot \left(\frac{du}{dx}\right)_w \approx \frac{u_4 - u_3}{h}$$

As the integrals are approximated over the half vol.

$$\cdot \int 2u dx \approx 2u_4 \frac{h}{2} = u_4 h$$

$$\cdot \int f(x) dx = f_4 \frac{h}{2}$$

$$-3 - \frac{(u_4 - u_3)}{h} - u_4 h = f_4 \frac{h}{2}$$

$$3u_3 - \frac{10u_4}{3} = \frac{8}{3} \quad - (Ka)$$

Final Solution.

(i) Dirichlet

	Exact sol ⁿ	FDM	FVM	FEM
(i) u_1 (at $x=0$)	0	0	0	0
(ii) u_2 (at $x=\frac{1}{3}$)	0.1111	0.1111	0.1077	0.1202
(iii) u_3 (at $x=\frac{2}{3}$)	-0.2222	-0.2222	-0.2274	-0.2105
(iv) u_4 (at $x=1$)	-1	-1	-1	-1

(ii) Neumann.

	Exact sol ⁿ	FDM	FVM	FEM
u_1	0	0	0	0
u_2	0.1111	0.1111	0.1190	0.1199
u_3	-0.2222	-0.2222	-0.1983	-0.1919
u_4	-1	-1	-0.9543	-0.9515